



A gamified AI movement assessment and feedback approach for university students in physical education: Effects on movement skills, learning engagement, and behavioral patterns

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Received: 10 April 2025 / Accepted: 26 January 2026 / Published online: 24 February 2026
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Abstract

University physical education courses often face challenges in sustaining students' learning engagement, particularly when practice tasks feel repetitive, progress is hard to perceive, and there is a lack of personalized feedback based on individual students' immediate performance. Although AI (Artificial Intelligence)-based movement assessment can help address the need for instant, individualized feedback, maintaining students' engagement over time remains a persistent challenge. In response, we developed a gamified AI movement assessment and feedback (G-AI-MAF) approach based on the ARCS motivational model (i.e., Attention, Relevance, Confidence, and Satisfaction), and adopted a quasi-experimental design to explore the impact of gamification mechanisms, such as levels, experience points, leaderboards, and player data, on student learning. The participants consisted of 80 students from two yogism classes at a university. One class served as the experimental group, adopting the G-AI-MAF approach, while the other class served as the control group, adopting the conventional AI movement assessment and feedback (C-AI-MAF) approach. The results showed that the G-AI-MAF approach significantly improved students' yogism movement skills performance and learning engagement. In terms of behavioral patterns, the experimental group took tests and checked results more frequently than the control group. These findings suggest that the gamification mechanism motivated students to engage in learning, significantly enhancing their active learning behaviors and the depth of system usage.

Keywords Movement assessment · ARCS model · Movement skills · Gamification · Physical education

1 Introduction

In physical education (PE), students' movement skills performance serves as a key indicator for assessing teaching effectiveness. Enhancing students' movement skills requires fostering their understanding of relevant theoretical knowledge and ensuring their proper execution of movements. Additionally, it is imperative to minimize incorrect practice to prevent potential sports injuries and declines in performance, ultimately achieving the educational goal of skill improvement. However, providing personalized guidance tailored to each student's individual needs presents significant challenges for teachers (Chao et al., 2021; Yan & Cardinal, 2013; Yan & FitzPatrick, 2016). As a consequence, scholars have introduced technology-assisted learning, which has successfully demonstrated its effectiveness in terms of supporting PE courses (Almusawi et al., 2021; Moon et al., 2023). In particular, the application of artificial intelligence (AI) has opened up more opportunities for technology-assisted learning and personalized guidance (Chen et al., 2022; Lai & Tu, 2024; Tu & Lu, 2025). For example, AI speech recognition technology can instantly translate and assist students in learning languages (Darvishi et al., 2022; Wang et al., 2023). Motion capture technology can analyze students' movement skills performance and provide personalized guidance based on individual learning progress and challenges (Hsia et al., 2024, 2025; Li et al., 2024; Song, 2024). These studies have indicated that the personalized feedback provided by AI can indeed help most students learn more efficiently. However, some students have reported difficulties in understanding the learning performance analysis provided by AI, suggesting that they require additional support to fully comprehend their issues (Liu et al., 2024; Song, 2024; Wang et al., 2023). To address this issue, some scholars have suggested providing a visualized learning analytics dashboard that presents key indicators through simple charts, helping students understand and reflect on their learning process (Lee & Kim, 2024). In line with this recommendation, this study presented assessment results in a visualized format to assist students in overcoming language barriers and in understanding movement skills guidance.

On the other hand, students' learning engagement and behavioral patterns are crucial factors for enhancing learning outcomes (Nair, 2022; Xiao & Hew, 2024). Therefore, promoting students' learning engagement and fostering more proactive learning behaviors are significant teaching challenges. In response, many scholars have advocated for the application of gamified learning strategies to motivate students to learn (Dehghanzadeh et al., 2024; Saleem et al., 2022). Gamified learning refers to integrating game elements such as leaderboards, badges, and challenges into non-game learning environments. Numerous studies have reported the effects of gamification on motivating students to learn, thus enhancing their learning participation (Chen & Chen, 2023; Nair, 2022; Xiao & Hew, 2024). The benefits of gamification in fostering more active learning engagement are also evident in students' learning behaviors. For example, well-designed gamified learning can encourage students to try different strategies to solve the challenges they face in

learning tasks (Chou et al., 2023; Shang et al., 2024). Additionally, gamification helps students focus more on the learning topic, effectively reducing distractions during the learning process (Chang et al., 2024). Overall, gamified learning contributes to motivating students' learning engagement and developing more proactive learning behaviors.

To ensure that gamification effectively enhances student learning, scholars have suggested designing gamified elements based on the ARCS motivational model (Camacho-Sánchez et al., 2023; Hao, 2023; Khan et al., 2022). Proposed by Keller (1987), the ARCS model comprises four dimensions: Attention, Relevance, Confidence, and Satisfaction. It emphasizes capturing students' attention, attracting their interest through novel and engaging content, and helping them focus on developing core competencies related to their learning goals. Enhancing relevance involves strengthening the connection between learning content and students' experiences, interests, and objectives to increase their perception of its practical value. The model also supports building students' confidence by setting clear goals, introducing progressively challenging tasks, and providing positive feedback, fostering their belief in their ability to succeed. Finally, it reinforces satisfaction by offering a sense of achievement when students complete learning tasks, leveraging both external rewards and intrinsic fulfillment to create a positive reinforcement loop (Khan et al., 2022; Quadir et al., 2023). Existing research has demonstrated that gamified learning based on the ARCS model can yield positive learning outcomes in higher education PE courses (e.g., Camacho-Sánchez et al., 2023). However, the integration of AI-supported real-time visual learning analytics to assist students in acquiring movement skills remains an unexplored area, with the specific implementation strategies and learning benefits requiring further investigation.

Therefore, this study adopted the ARCS motivational model as the foundation for designing a gamified AI movement assessment and feedback (G-AI-MAF) approach. The approach incorporated gamification mechanisms such as levels, experience points, leaderboards, and player profiles to enhance student engagement and foster more proactive and efficient learning behaviors. Additionally, it utilized visualized learning analytics to help students overcome difficulties in interpreting AI-generated skill assessments and feedback. By integrating these elements, the study sought to enhance the enjoyment of yogism courses for students, ultimately improving their learning experience and skill acquisition. The following research questions were addressed in this study:

- (1) Does the G-AI-MAF approach lead to a difference in students' movement skills performance compared to the conventional AI movement assessment and feedback (C-AI-MAF) approach?
- (2) Does the G-AI-MAF approach lead to a difference in students' learning engagement compared to the C-AI-MAF approach?
- (3) Does the G-AI-MAF approach lead to a difference in students' learning behaviors compared to the C-AI-MAF approach?

2 Literature review

2.1 Learning challenges of students and the support provided by digital technologies in PE courses

The primary goal of university PE courses is to develop students' movement skills, enabling them to apply these skills in their daily lives. However, the large-class format commonly adopted in PE courses often makes individualized instruction challenging, particularly in teaching movement skills. Chiang et al. (2019) found that personalized instruction tailored to specific skills could significantly enhance students' performance, although it required more teaching resources.

The application of digital technology offers a potential solution, demonstrating its ability to enhance instructional effectiveness. For example, Moon et al. (2023) developed a real-time virtual PE class observation system to support the quality of instruction in remote teaching settings. The integration of wearable technology has also been shown to improve students' movement skills performance and health behaviors (Almusawi et al., 2021). Additionally, Chen (2024) indicated that new media teaching models incorporating big data analytics could significantly enhance students' teamwork skills. Furthermore, digital teaching approaches have been shown to significantly influence students' motivation for sports participation (Botagariyev et al., 2024). These studies highlight the important role of technology-assisted instruction in improving the effectiveness of PE courses and enhancing students' overall learning experiences.

In this regard, AI-assisted learning has emerged as a potential solution. AI can provide instant and personalized feedback based on individual students' practicing performance (Guo & Li, 2021; Hsia et al., 2024; Song, 2024). As a result, designing intelligent and personalized AI learning tools is a critical issue for achieving more effective cross-cultural PE education and fostering students' skills development.

2.2 AI-Assisted learning

AI-assisted learning refers to the use of AI technologies to support the teaching and learning process. These technologies include functions such as speech recognition, image analysis, and automated assessments, which can provide personalized learning recommendations, real-time feedback, and resource allocation (Chang & Hwang, 2024; Darvishi et al., 2022). For instance, Guo et al. (2024) developed an AI-supported system, EvalMate, to offer high-quality peer feedback, thereby significantly enhancing students' feedback literacy and learning motivation.

The advantage of AI lies in its ability to adapt to the diverse needs of learners. For example, Liu et al. (2024) showed that generative AI technology could help students complete multimodal academic creations by integrating image generation and text editing tools, thereby enhancing students' creative abilities and academic performance. Similarly, Wang et al. (2023) found that AI systems, using deep learning methods, could support learners in achieving better learning outcomes, particularly in language learning. These findings highlight the potential of AI technology to enhance learning efficiency and personalized instruction. Additionally, AI can improve the

transparency and effectiveness of the teaching process through data analysis techniques. For instance, Rad et al. (2024) used AI to provide real-time feedback, which improved students' writing skills and learning engagement. Also, the RiPPLE system developed by Darvishi et al. (2022) optimized the peer assessment process using AI, enhancing trust among peers and the quality of assessment. These applications demonstrate the positive impact of AI on collaboration and feedback mechanisms in the learning process.

The application of AI technology in PE courses is becoming increasingly widespread, particularly in overcoming the challenges of personalized instruction and skills assessment, demonstrating significant value. For instance, Yangsheng (2021) developed a sports performance prediction and evaluation system based on recurrent neural network (RNN) technology, which integrated classroom and extracurricular data to provide comprehensive assessments, significantly enhancing the precision and objectivity of teaching. Additionally, Song (2024) designed an intelligent tennis teaching system that effectively improved teaching quality and enhanced students' learning efficiency. In terms of functional movement training, Guo and Li (2021) explored an AI-supported intelligent PE teaching system that utilized data analysis to provide personalized teaching strategies, thereby increasing students' engagement and learning outcomes. This AI-based teaching model not only overcame the time and space limitations of conventional instruction, but also fostered autonomous learning and skills mastery. Moreover, Hsia et al. (2024) found that an AI-assisted automatic yogism assessment system could provide real-time feedback, helping students improve their movement skills and increasing their learning motivation. The application of AI technology also supports immersive experiences in PE education. For example, Li et al. (2024) combined AI with metaverse technology, using 360-degree VR videos to optimize soccer teaching, thereby improving students' learning experiences and outcomes. These studies demonstrate that AI technology provides innovative solutions for PE courses, enhancing teaching efficiency and creating more diverse learning environments.

However, insufficient student engagement remains a significant challenge in PE courses, making it critical to find effective ways to enhance participation. Scholars have suggested that incorporating gamification strategies can significantly improve student engagement (Feng et al., 2024). For example, designing engaging interactive experiences and competitive mechanisms can promote active learning.

2.3 Gamified learning mechanisms

Gamification refers to the application of game elements (e.g., leaderboards, badges, and challenges) in non-game contexts to enhance participant engagement and behavioral performance (Nair, 2022). The core of gamification lies in increasing the appeal and interactivity of the learning process through incentive mechanisms. For example, Xiao and Hew (2024) found that tangible rewards significantly enhanced students' intrinsic motivation and behavioral engagement, particularly in digital learning environments. Qiao et al. (2023) discovered that hybrid gamification designs had a notable advantage in promoting self-regulated learning and increasing course satisfaction. Furthermore, Chen and Chen (2023) proposed a real-time perspective

comparison system combined with gamification mechanisms (e.g., leaderboards and instant feedback), which significantly improved students' discussion effectiveness and their ability to consider diverse perspectives.

Research has shown that integrating the ARCS model with gamified learning leads to more significant effects. The ARCS motivational model, proposed by Keller (1987), aims to enhance both learning motivation and outcomes. It consists of four core dimensions: Attention, Relevance, Confidence, and Satisfaction, which provide a foundation for designing motivating instructional activities. For example, Hao (2023) explored a digital teaching game based on the ARCS learning motivation theory, and found that the experimental group with gamification showed significant improvement in learning motivation across all four dimensions. Lin and Hou (2024) applied the ARCS motivation model to an augmented reality (AR) educational board game, examining its impact on students' learning motivation. The results showed significant improvements in Attention and Satisfaction. Furthermore, Camacho-Sánchez et al. (2023) demonstrated that ARCS-based gamified digital learning significantly enhanced students' Attention, Relevance, Confidence, and Satisfaction in PE courses, further boosting their academic performance.

In PE courses, gamified learning models not only stimulate students' interest in sports, but also promote skills development and behavioral improvement. For example, Quintas et al. (2020) employed a gamified design using exergames, which significantly enhanced elementary school students' fulfillment of their basic psychological needs and rhythmic movement skills. Additionally, Hsia et al. (2025) indicated that a gamified intelligent-assisted teaching system based on self-determination theory (SDT) improved students' engagement in yogism skills learning and their skills mastery. Studies have also pointed out that gamified PE instruction can integrate digital and physical tools, creating diverse learning contexts. For instance, Wang et al. (2024) discovered that leaderboard designs could be personalized according to students' competitiveness traits to optimize their learning outcomes. Furthermore, gamified elements motivated students to engage in more physical activities, thereby enhancing their learning effectiveness.

The integration of gamification and AI systems further expands educational innovation. In physical education contexts, only a limited number of empirical studies have demonstrated the effectiveness of this combination. For example, Hsia et al. (2025) reported the effectiveness of using a gamified intelligent tutoring and feedback system for yoga skill learning from the perspective of SDT; Gao et al. (2025) also employed a gamified intervention with AI-supported personalized recommendations in physical activities from the perspective of SDT. In addition, Park et al. (2025) introduced a digital gamified activity program combining AI-driven personalized exercise prescriptions for elementary school students' health-related physical fitness. However, these studies did not focus on learning engagement issues from the perspective of the ARCS Motivational Model using AI-based movement detection mechanisms, and nor did they analyze students' behavioral patterns. The innovation of the present study lies in adopting the ARCS model as the core design framework for the proposed G-AI-MAF approach, integrating the AI system's real-time visualized feedback with ARCS-oriented gamification mechanisms. This approach aims to

enhance learners' attention, build learning confidence and satisfaction, and thereby foster students' sustained engagement and proactive learning behaviors.

3 The G-AI-MAF approach

For this study, we employed C++ to develop a G-AI-MAF approach aimed at enhancing student engagement through novel and engaging gamification mechanisms. Additionally, the approach sought to support students in understanding AI-generated personalized feedback via a visualized learning analytics dashboard. As shown in Fig. 1, the system consists of two primary modules: the gamified visual interactive module and the AI recognition and analysis module.

First, the AI recognition and analysis module employed OpenPose to capture key data from students' yogism postures, and generate skeleton diagrams. A deep learning model was then used for accuracy analysis. This study adopted Hsia et al.'s (2025) knowledge model to compute key angles of detected yogism postures for performance evaluation. The model demonstrated a high level of agreement with expert ratings, with a Kendall's ω value ranging from 0.708 to 0.818 ($p < .00$), indicating its reliability in assessing yogism postures with expert-level precision. In terms of predictive performance, the model's accuracy was 0.97,

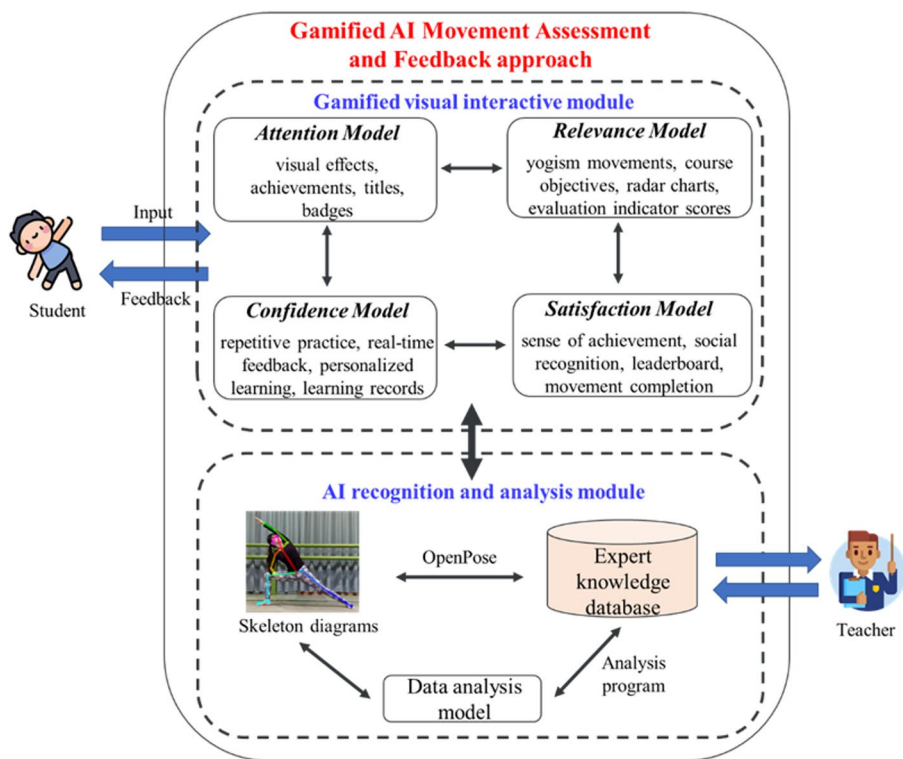


Fig. 1 System structure

and recall, precision, and F1-score were 0.96, demonstrating exceptional precision in yogism posture recognition. Based on evaluation indicators defined by experts, the analytic results were then presented as visualized feedback, such as radar charts and bar graphs. By utilizing graphical representations, tables, and colors, the system enabled students to understand their performance and identify areas for improvement.

Second, the design of the gamified visual interactive module gamification mechanisms was based on Keller's (1987) ARCS motivation model. The details are provided as follows.

3.1 Attention model

To attract students' attention, the system featured a clear and intuitive user interface along with gamified interactive competition elements to sustain their engagement. Additionally, it incorporated personalized virtual avatars, allowing students to customize their character's appearance, background, and nickname. Based on their scores, students could earn titles, achievements, and badges (see Fig. 2). These gamification mechanisms provided visual stimuli and real-time interactive feedback, effectively capturing students' attention and fostering a stimulating, curiosity-driven learning experience.

3.2 Relevance model

The system integrated learning objectives, students' interests, and individual learning needs to enhance the relevance of the content. For example, students could select yogism movements they wished to assess based on their interests and needs, watch demonstrations, and receive key explanations of those movements. They were also allowed to submit practice photos to obtain instant advice, including skeleton diagrams, radar chart analysis, and indicator scores (see Fig. 3). Through the system's specific guidance and feedback, students gained a clearer understand-

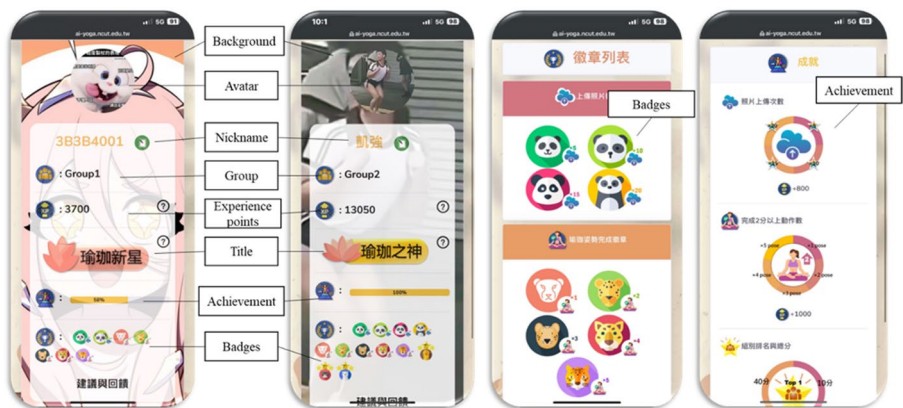


Fig. 2 Gamified personal interfaces



Fig. 3 Interfaces for movement monitoring and feedback

ing of how to improve their performance. At the same time, this strengthened the connection between the practice process and specific learning outcomes, further enhancing the meaning and relevance of their learning.

3.3 Confidence model

The system provided a personalized learning environment to help students gradually build their self-confidence. For instance, students could review learning resources and practice at their own pace. After each assessment, the system offered real-time feedback, allowing students to intuitively understand their progress in mastering the skills. Moreover, the system provided students with a learning record to review feedback from previous assessments (see Fig. 4). By viewing their learning records, students could connect their efforts with their progress, have more opportunities to practice based on their needs, and gradually improve their performance, thereby boosting their learning confidence.

3.4 Satisfaction model

To enhance students' learning satisfaction, the system included a series of incentive mechanisms that provided a sense of achievement during the learning process. For example, after completing specific tasks, students unlocked achievement badges or level upgrades, which they could view on their personal homepage. Additionally, students' learning performance was converted into group scores. Through collaboration with group members, they could collectively rise on the group leaderboard, fulfilling their need for social recognition and satisfaction (see Fig. 5). Finally, the system provided visual feedback on the degree of movement completion, allowing students to understand the extent to which each movement was mastered. This helped them see the results of their efforts, further enhancing their learning satisfaction.

Fig. 4 Interfaces for learning records



Fig. 5 Interfaces for the leaderboard and degree of movement completion

4 Research design

This study aimed to verify the impact of the G-AI-MAF approach on learners' movement skills performance. It was applied in a yogism course to collect relevant data. Students were divided into an experimental group and a control group. After completing the learning activities, we compared the effects of the different learning approaches on students' learning outcomes, and explored the underlying reasons for any differences.

4.1 Participants

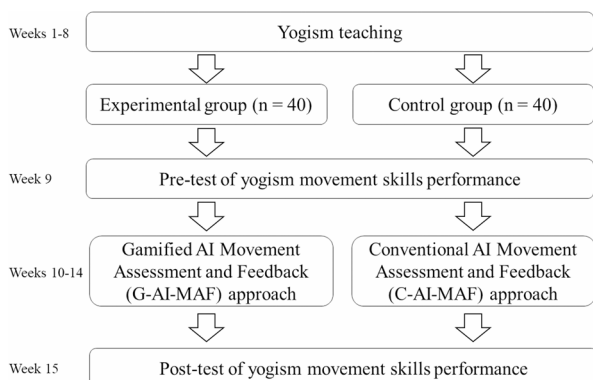
This study adopted a quasi-experimental design. The participants were two classes of university students from a university of science and technology in central Taiwan. All of them were from non-sport-related academic discipline, and their ages ranged from 19 to 20 years. Most were beginners who had never been exposed to yogism before.

One class ($n=40$) was assigned as the experimental group using the G-AI-MAF approach, and the other ($n=40$) served as the control group using the C-AI-MAF approach. To mitigate potential selection bias associated with this class-level assignment, a pretest was conducted. The homogeneity test (Sect. 5.1) indicated no significant differences in the two groups' initial skill levels. Moreover, all of the students were taught by the same teacher with over 10 years of yogism teaching experience.

4.2 Experimental procedure

The experiment was conducted in two yogism classes of 100 min per week. The experimental procedure is shown in Fig. 6. During weeks 1–8, the teacher introduced the teaching content, test items, and evaluation criteria, and assigned students to their groups for the class activities. The course content began with warm-up and protective exercises to teach students how to perform movements safely, followed by the instruction of a set of essential yogism movements. In the 9th week, a pretest of yogism movement performance was conducted.

Fig. 6 Experimental procedure



The teaching intervention was conducted during weeks 10 to 14, with each week focusing on an advanced yogism movement. By the end of the 5 weeks, students would learn a set of advanced yogism movements. In the learning process, the teacher first introduced, explained, and demonstrated the movements, followed by student practice. Both groups of students employed the AI movement assessment and feedback system for learning. They worked in small groups, took photos of their movements, and uploaded them to the system. The AI system would then evaluate the accuracy of the movements and provide corresponding feedback and scores, offering students a basis for learning and improvement (see Fig. 7). What differentiated the two groups was that the experimental group used the approach integrated with gamification mechanisms. After uploading their movements, students in the experimental group could view their rankings relative to the entire class and earn rewards (e.g., badges and titles) and achievements. Additionally, students could view their group rankings, which fostered competition and engagement through gamification. In contrast, the control group only received feedback on their performance from the AI system without any game elements. A posttest of yogism movement skills performance was administered in the 15th week.

4.3 Measuring tools

4.3.1 Yogism movement skills performance

This study adopted the yogism movement skills rubric developed by Hsia et al. (2024) to assess students' yogism performance. The rubric categorized movement skills into four levels, with a score range from 1 to 4. A score of 4 indicates perfect

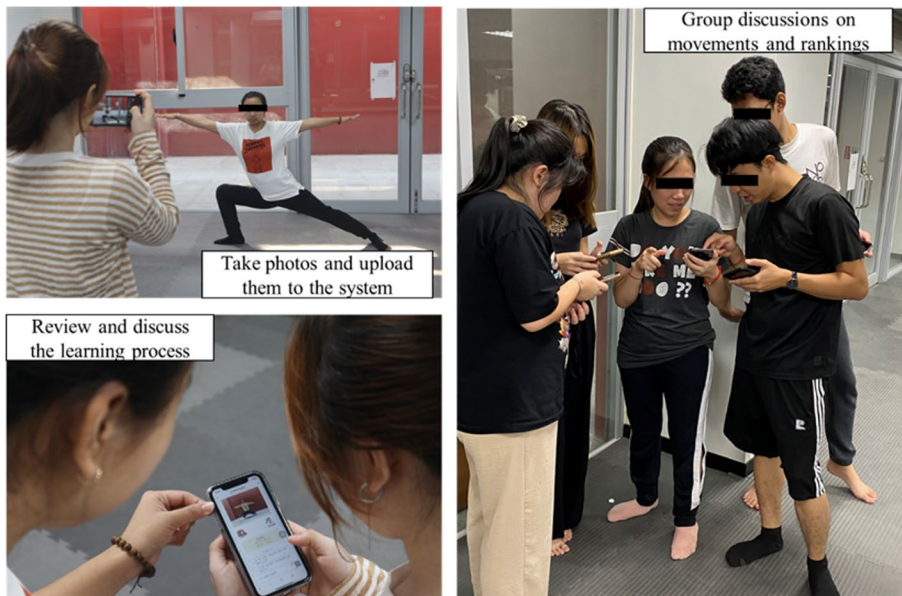


Fig. 7 Examples of students using the system

execution of movements with no noticeable deficiencies, including accuracy, range of motion, stability, and strength. A score of 1 reflects significant errors in execution, such as severe slouching or major technical mistakes. Two teachers, each with over 5 years of yogism teaching experience and certified as yogism instructors, were recruited to grade the students' videos. The results showed high consistency between the two raters, with a Spearman's rank correlation coefficient of 0.79 ($p < .001$). The average scores of the raters were adopted as the final scores for the students' yogism movement skills test.

4.3.2 Learning engagement

This study adapted the learning engagement formula proposed by Hwang et al. (2007) to fit its context. The formula for calculating learning engagement is $LE(S_i) = (T(S_i)/MaxT + R(S_i)/MaxR)/2$, where $LE(S_i)$ represents the learning engagement score of a student S_i , $T(S_i)$ is the number of times student S_i logs into the system, and $MaxT$ is the highest login frequency among peers. $R(S_i)$ is the number of photos uploaded by student S_i , and $MaxR$ is the highest number of photo uploads among peers. The learning engagement score ranges from 0 to 1, with higher values indicating greater student engagement. This formula was used in this study to evaluate students' engagement in the learning activities.

4.3.3 Behavioral patterns

This study adopted the game-based learning behavior coding scheme proposed by Yang et al. (2018) and applied it to its learning context. The revised coding scheme is outlined in Table 1.

4.4 Data analysis

In terms of yogism skills performance, this study used an ANCOVA to control the effect of students' prior competences on the experimental results. After adjusting the pretest scores, the posttest scores of the two groups were compared to determine whether the teaching intervention led to significant differences. The significance level was set at $p < .05$. If a significant difference was found, the η^2 value was calculated to assess the effect size.

Regarding learning engagement, after the teaching intervention, the frequency of logins and the number of practice photo uploads were calculated based on data recorded in the system's backend. The data were analyzed using an independent samples t test to compare the two groups' learning engagement. The significance level was set at $p < .05$. If a significant difference was found, the Cohen's d value was calculated to assess the effect size.

As for learning behaviors, the system automatically recorded students' actions, and lag sequential analysis was conducted using the GSEQ software to explore behavioral patterns within the learning environment. Z-scores were employed for testing, and adjusted residuals tables were generated to analyze these behavioral patterns. A

Table 1 Coding scheme for students' learning behaviors

Code	Definition	Description
L	Log in	Log into the system
W	Take the weekly unit test	Take the movement skills test related to the current course content
T	Take the individual test	Upload practice photos and take movement skills tests not related to this week's content
R	View the latest test results	View the results of the latest movement skills test
P	Review previous test results	Review completed movement skills test results
E	View teaching content	Display the teacher's demonstration and explanation
D	Check the dashboard	View the dashboard with an overview of movement completion degree

Table 2 ANCOVA results of yogism movement skills performance

Group	<i>N</i>	Mean	SD	Adjusted Mean	SE	<i>F</i>	η^2
Experimental group	40	2.46	0.33	2.47	0.05	19.93***	0.21
Control group	40	2.19	0.25	2.18	0.05		

*** $p < .001$

Z-score greater than 1.96 (Bakeman & Gottman, 1997) was considered significant for identifying meaningful behavioral patterns.

5 Results

5.1 Yogism movement skills performance

To exclude the impact of pretest scores on the results, the homogeneity of regression coefficients was first tested. The results indicated that the F -value did not reach significance ($F=3.09$, $p=.08 > .05$), suggesting that the pretest scores of the two groups were homogeneous. Therefore, ANCOVA could be performed.

Table 2 shows that the experimental group demonstrated significantly better learning performance than the control group ($F=19.93$, $p=.000 < .001$), with a large effect size ($\eta^2 = 0.21$). The adjusted mean for the experimental group was 2.47, while that for the control group was 2.18. This indicated that the G-AI-MAF approach led to superior yogism movement skills performance.

5.2 Learning engagement

During the 5-week teaching intervention, 785 login records and 824 upload records were collected from students. As shown in Table 3, an independent samples t test revealed that the experimental group had significantly higher learning engagement than the control group ($t=3.33$, $p=.001 < .01$), with a small effect size ($d=0.13$).

Table 3 Independent *t*-test results of learning engagement

Group	<i>N</i>	Mean	SD	<i>t</i>	<i>d</i>
Experimental group	40	0.51	0.16	3.33**	0.13
Control group	40	0.42	0.09		

***p* < .01**Table 4** Adjusted residuals table of the experimental group

Given	L	W	T	R	P	E	D
L	2.5*	7.98*	−1.37	−10.06	−0.96	−0.8	1.24
W	−4.08	−6.09	−4.56	32.82*	−9.93	−8.01	−4.8
T	−1.89	−3.22	7.74*	8.74*	−2.56	−2.5	−2.44
R	−3.35	12.86*	3.71*	−9.7	−8.52	8.66*	−2.07
P	−0.37	−9.2	−1.08	−10.85	13.54*	1.3	7.66*
E	3.04*	1.55	0.96	−6.42	0.79	5.3*	−3.05
D	5.43*	−4.25	0.27	−8.43	8.98*	−2.84	2.18*

Table 5 Adjusted residuals table of the control group

Given:	L	W	T	R	P	E	D
L	−0.54	12.73*	3.2*	−11.35	−3.9	−0.08	−0.28
W	−7.21	−9.47	−2.56	27.44*	−2.98	−7.61	−1.95
T	−1.85	−3.74	1.75	7.37*	−1.52	−2.86	1.38
R	6.4*	−0.04	−1.2	−7.54	−3.49	7.67*	−1.95
P	0.42	−3	−0.42	−5.11	6.85*	2.04*	1.79
E	2.9*	0.7	−0.68	−6.86	−0.17	2.78*	2.21*
D	1.1	−1.11	0.55	−6.15	9.35*	−1.1	0.65

5.3 Analysis of behavioral patterns

The system developed in this study automatically recorded 4,160 learners' behaviors. These behaviors included logging in (L), taking the weekly unit test (W), taking the individual test (T), viewing the latest test results (R), reviewing previous test results (P), viewing teaching content (E), and checking the dashboard (D). Subsequently, lag sequential analysis was performed, and Z-scores were calculated. Tables 4 and 5 present the adjusted residuals tables. Figure 8 illustrates the behavioral transition diagrams for both groups of students.

Both groups of students shared similarities and differences in their learning behaviors. In terms of similarities, the types of learning activities were consistent, and both groups demonstrated a core behavioral cycle: starting with logging in (L), followed by taking tests (T), and then checking the results (R).

In terms of differences, these were primarily reflected in the direction and frequency of behavioral transitions. First, the experimental group demonstrated bi-directional interactions for key behaviors, such as the transitions between W-R, T-R, and P-D, while the control group showed one-way behavior. This indicated that the experimental group more frequently used the following functions: repeatedly checking individual test results, monitoring overall learning performance, and retaking tests. Second, the behaviors T (individual test) and D (dashboard) showed significant cycles in the experimental group.

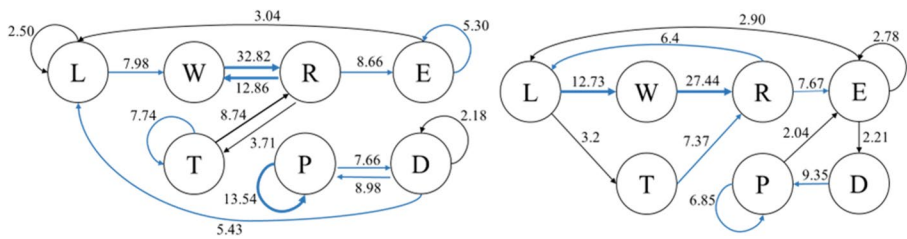


Fig. 8 The learning behavioral patterns of students (left: the experimental group; right: the control group)

As for the Z-scores, the experimental group had the highest Z-score for the behavior $W \rightarrow R$ (32.82), demonstrating their strong focus on receiving immediate feedback after taking the weekly test. Similarly, for the behaviors $T \leftrightarrow R$ and $P \leftrightarrow D$, the experimental group significantly outperformed the control group, demonstrating that students more frequently switched between test results and learning data. In contrast, the control group's behavioral patterns were more simplified, with less frequent utilization of feedback from test results.

These differences were primarily reflected in more frequent transitions related to test-taking and result review in the experimental group. The experimental group demonstrated higher learning efficiency and more diverse behavioral patterns, particularly in terms of viewing test results and utilizing system functions more actively. In contrast, the control group had more simplified behaviors and limited use of system functions, leading to relatively lower learning engagement.

6 Discussion and conclusions

This study proposed the G-AI-MAF approach integrated with the ARCS motivational model. It offered students real-time personalized visual feedback and guidance for acquiring movement skills within a gamified environment.

The results indicated that the experimental group demonstrated significantly better yogism movement skills performance compared to the control group, confirming the effectiveness and practicality of the G-AI-MAF approach developed in this study. By integrating gamification strategies into the AI system, it provided engaging tasks and real-time interactive competitive rankings, which increased students' frequency and persistence in actively participating in practice. Previous studies have also pointed out that learning systems incorporating game elements can stimulate students' intrinsic competitive awareness, enhancing their engagement and sense of accomplishment during the learning process, thereby effectively promoting skill acquisition and optimizing overall learning outcomes (Feng et al., 2024; Wang et al., 2024). This study utilized AI technology combined with the ARCS model. The system could instantly analyze students' uploaded yogism movements, perform precise comparisons and error detection based on each student's movement accuracy and individual skill level, and generate personalized, visual, real-time feedback. Through the gamified mechanism with immediate and specific feedback, students could clearly identify

shortcomings in their movements and make adjustments and corrections right away, significantly improving their learning efficiency and performance in yogism movement skills (Guo & Li, 2021; Hsia et al., 2024, 2025; Song, 2024). Therefore, the G-AI-MAF approach introduced in this study not only overcame the limitations of conventional yogism courses that lack real-time feedback, but also successfully promoted students' yogism skills through enhanced engagement and interactivity, providing a new direction and significant practical value for movement skills learning.

In terms of learning engagement, the experimental group adopting the G-AI-MAF approach demonstrated significantly higher learning engagement than the C-AI-MAF group. This indicated that the G-AI-MAF approach had a significant positive impact on students' learning engagement. Through this approach, students could receive real-time feedback on their movement skills performance. The gamified incentive mechanism further stimulated their learning motivation and willingness to engage in learning activities. This finding was consistent with previous research, suggesting that AI systems incorporating game elements can effectively enhance students' learning motivation and learning engagement (Guo & Li, 2021; Hsia et al., 2024).

In terms of behavioral patterns, the G-AI-MAF group demonstrated more repeated behaviors, such as viewing test results after taking the weekly unit test (W-R), viewing test results after taking the individual test (T-R), and checking the dashboard after reviewing previous test results (P-D). This suggested that the experimental group was more proactive and willing to engage in learning. Furthermore, the experimental group had the highest Z-score for the behavior W-R. This implied that students were particularly focused on receiving immediate feedback after completing the test, which contributed to improving their learning outcomes (Shang et al., 2024). The game elements effectively increased students' interest in learning, helping them to take tests more frequently and to actively check the results (Chou et al., 2023; Shang et al., 2024). Overall, the gamified design significantly enhanced students' active learning behaviors and the depth of system usage, providing important insights for future learning system design and teaching strategies.

This study integrated the ARCS motivational model, which includes Attention, Relevance, Confidence, and Satisfaction, to ensure that the system effectively met the diverse psychological needs of students while offering gamified interactions. Students were more likely to repeatedly use the system due to specific applications of the ARCS model, such as using visual feedback to capture their attention, designing tasks that align with their life experiences to increase relevance, enhancing their confidence through immediate reward mechanisms, and boosting learning satisfaction through a sense of accomplishment upon task completion. This, in turn, promoted their active learning and improved their movement skills, leading to deeper learning engagement. The results of this study align with those of Camacho-Sánchez et al. (2023), Hao (2023), and Lin and Hou (2024), which indicated that integrating the ARCS motivational model into gamified learning systems could significantly enhance students' engagement and active participation in learning. Therefore, this study not only reinforced previous theoretical perspectives, but also demonstrated the effectiveness and feasibility of combining gamified AI systems with the ARCS motivational model in the field of movement skills learning.

Despite its contributions, the limitations of the present study should also be acknowledged. That is, the sample comprised students from a single institution in a specified course taught by the same instructor. Although this design was intended to control for teacher variability and ensure internal validity, it potentially limits the representativeness and external validity of the findings. It is suggested that future research validate the G-AI-MAF approach across diverse institutions and encompass a broader range of nationalities to enhance generalizability. Furthermore, as this study focused solely on yogism, future studies could extend the application to various types of physical activities and explore the integration of different game elements with AI technologies to further investigate their effectiveness.

Authors' contributions All authors contributed to the study conception and design. Formal analysis, Investigation, Data Curation, Original Draft and Writing were performed by Yu-Hsien Huang. Methodology, Formal analysis, Validation, Original Draft, Writing, Review and Editing were performed by Lu-Ho Hsia. Conceptualization, Methodology, Writing, Review and Editing were performed Gwo-Jen Hwang. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding Open access funding provided by National Taiwan University of Science and Technology. This study is supported in part by the Ministry of Science and Technology of Taiwan under contract numbers NSTC 114-2628-H-167-002-MY3 and NSTC 112-2410-H-011-012-MY3. The study is also supported by the “Empower Vocational Education Research Center” of National Taiwan University of Science and Technology (NTUST) from the Featured Areas Research Center Program within the framework of the Higher Education Sprout Project by the Ministry of Education (MOE) in Taiwan.

Data availability The data and materials are available upon request to the corresponding author.

Code Availability Not applicable.

Declarations

Ethics approval The ethical requirements for research in this selected university were followed.

Consent for publication The publication of this study has been approved by all authors.

Consent to participate The participants all agreed to take part in this study.

Conflicts of interest/Competing interests There is no potential conflict of interest in this study.

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